

National College of Ireland

H9IAPA: Intelligent Agents and Process Automation

MSCAIJAN26I MSCAIBUSJAN26I

Release Date: 29 June 2026 at 09:00

Submission Deadline: 31 July 2026 at 23:55

Continuous Assessment (CA) Type: Individual Assignment

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Weight: The assignment will be marked out of 100. This CA is worth **100%** of the overall marks for this module.

Instructions: This CA is individual work.

Any AI-generated content (text, images, figures, code, etc.) used must be disclosed. AI tools **must not be used to fabricate or manipulate** data, code, figures, text or other results. Misuse of AI in this way constitutes academic misconduct and will result in failure.

Submission Details:

The report (PDF) implemented solution with any accompanying data, code, and presentation must be submitted to Moodle before the deadline. The submitted report must be no more than 10 pages in length. Use a single column layout document. The font size for the body of the text should be 12-point Times New Roman. Include student name, student ID, and course name at the top of the first page. Make sure to acknowledge the original sources for your investigation as appropriate.

Late Submissions: Late submissions will be subject to a late submission penalty except where a student has applied and been approved for an extension through NCI360.

TURNITIN: All report submissions will be electronically screened for evidence of academic misconduct (i.e., plagiarism and collusion)

Description:

As part of your final assessment for the module, you are required to propose and execute a project that involves identifying a process, modelling it using Business Process Modelling Notation (BPMN), investigating tasks within the process for their automation potential, and proposing automation solutions.

The project must include two distinct automation solutions: one implemented using Robotic Process Automation (RPA) and one using agentic (or AI-driven) automation. These solutions may address different tasks within the same overall process or may involve two separate processes. Where two separate processes are selected, both must be modelled using BPMN.

Project Tasks**1. Process Identification:**

Identify a process in a domain such as Banking, IT, HR, Retail, Insurance, Hospitality, or any other domain where automation could add value. Provide a brief overview of the chosen process and explain its relevance to the selected domain.

2. Process Modelling:

Use BPMN to create a detailed model of the identified process. Include all relevant tasks, decision points, and subprocesses. Show both the As-Is and To-Be scenarios. If your project involves automating two separate processes, both processes must be modelled using BPMN.

3. Automation Potential Analysis:

Analyse the tasks within the modelled process(es) and identify those with potential for automation. For each task, briefly justify whether it is suitable for automation and why.

4. Automation Proposals:

Propose two distinct automation solutions:

- **RPA-based automation:** Select one or more tasks suitable for RPA and provide a detailed proposal including the approach, tools/technologies required, expected benefits, and potential challenges.
- **Agentic or AI-driven automation:** Select one or more tasks suitable for intelligent automation (e.g., using machine learning, natural language processing, or cognitive automation). Provide a proposal similar to the RPA solution.

These solutions may target different tasks within the same process or tasks from two separate processes.

5. Solution Implementation:

Implement and demonstrate **both** of the proposed automation solutions using a tool of your choice (e.g., Python, UiPath). Include sample data, code (if applicable), and necessary documentation. If using external APIs, ensure any required API keys are included in the implementation.

Deliverables:

1. **Report:** A well-documented report explaining the tasks involved.
2. **Solution Implementation:** A UiPath/Python implementation of the solutions.
3. **Video Presentation and Slides:** A 7-minute video explaining the business process identified, BPM model, the task considered for automation, and the solution developed.

Using the following scale, each task in this section will be awarded a percentage of the total mark.

Grade Criterion	H1 (> 70%)	H2.1 (> 60%)	H2.2 (> 50%)	Pass (> 40%)	Fail (< 40%)
Process Identification and Description: 10%	Interesting and challenging process identified and well discussed.	Reasonable process identified and well described.	Reasonable process and mostly described.	Process partially described	Process description not clear
Process Modelling: 15%	The process model is exceptionally clear and comprehensive, including all relevant elements such as events, tasks, subprocesses, decision points, and flows.	All stages of the process model are clear and mostly complete, with minor omissions or ambiguities that do not significantly impact understanding.	The process model is adequately clear but lacks some detail or completeness, making it slightly challenging to follow in certain areas.	The process model is somewhat unclear or incomplete, requiring additional effort to understand the flow of the process.	The process model is unclear and significantly incomplete, making it difficult to discern the sequence of tasks or subprocesses.
Automation Potential Analysis: 10%	Comprehensive evaluation of automation possibility of all tasks involved, addressing both short-term and long-term implications and risk assessment.	Comprehensive evaluation of automation possibility of most tasks involved, addressing both short-term and long-term implications.	Good evaluation of automation possibility of most tasks involved.	Basic evaluation of automation possibility of some tasks involved.	Poor, incomplete evaluation of few of the tasks involved.
Automation Proposal: 15%	The proposal clearly articulates a sophisticated approach for automation, integrating advanced tools or technologies that are highly suitable for the task. Thoughtful exploration of intelligent automation possibilities.	A well-defined approach is presented with appropriate tools or technologies selected, demonstrating a strong alignment with the task requirements. Thoughtful exploration of intelligent automation possibilities.	The proposal outlines a viable automation approach, with relevant tools or technologies identified to support the task.	An acceptable approach is described, although it lacks depth or clarity in tool selection or suitability for the task.	The automation approach and choice of tools are unclear or insufficiently detailed, hindering understanding and feasibility assessment.
Solution Implementation: 30%	Exceptionally clear and very accurate solutions developed around the proposal.	Mostly clear and accurate solutions developed around the proposal.	Developed solutions lacks reasonable accuracy.	Developed solutions do not work as expected.	No solutions developed.
Report: 10%	Well written, with no (large) language errors. All figures are well-conceived and easy to read. The report does not exceed the length limits. References are complete, appropriately and correctly used.	The report has a few language and/or style errors. The figures are well presented. Format and length limits are adhered to. References are complete and correctly used.	The report is readable with some language and/or style errors. Some figures may be hard to read or presented in a suboptimal manner. References are mostly complete and correctly used.	The report is readable with many language and/or style errors. Most figures are not clear or easy to read. References are few and/or mostly incomplete.	Littered with typos, and/or poor use of English. The figures are poor and hard to read. References (if any) are probably incomplete and poorly used.
Video Presentation: 10%	The presentation clearly outlined the project argument. Slides were error-free and logically presented. Detailed solution demonstration exists.	The presentation somewhat clearly outlined the project argument. Slides were somewhat error-free and somewhat logically presented. Detailed solution demonstration exists.	The presentation outlined the project argument. Slides were mostly error-free and mostly logically presented. Solution demonstration exists.	The presentation provided a limited outline of the project argument. Slides were not error-free and not logically presented. No solution demonstration.	The presentation is unorganised and unclear. No solution demonstration.